AI ETHICS

Algorithmic Bias and Relativity in AI

By Ken Ito

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Hello, dear viewers, this is Ken Ito from Tokyo, and today I'd like to discuss one of the most important recent topics in AI ethics that is algorithmic bias. I hope my talk will be a little bit beyond your expectations. Here we go. What does algorithmic bias mean today? This means social unfairness and inequality produced by AI. Timnit Gebru, a computer scientist is one of the leading researchers in this field, one in Ethiopia. And after sitting at Stanford, she has worked on algorithmic bias, especially focusing on race and gender at Google. On 2020, she published a paper which showed the possible errors in a large scale natural language AI system, the issues of energy consumption for the machine learning caused by the system and the issues of environmental burdens it possibly brings. Over her publication, she eventually left from Google, and it became a scandal. But this is not the main interest of my talk today. My main interest lies in the fundamentals of AI ethics, then, again, what is algorithmic bias? Briefly speaking, the answer is AI would have prejudice. In any human society there exists various kinds of discrimination, hate, and wrong recognition. These social biases were embedded in AI, and the forms of datasets, and AI itself imitates social bias and starts to think with prejudice. This is called algorithmic bias. Such a potential problem had been pointed out already in the year 2012. Generally speaking, however, it's safe to say that it was at the Asilomar Conference on artificial intelligence in January 2017 that the AI problem of bias was widely recognized, noticed, and after that, the problem in question has become widely known all over the world. Here is an example of the AI bias problem I'm talking about. Suppose that there is a data set, which shows the high rate of crime by social minority, for example, black people, and an AI system which learns such a bias that might show as its result, a certain high potential rate of crime by black people. It's also well-known that the facial distinction rate of the white male is much higher than that of black female people. I argue that unbalanced data will eventually give rise to this biased system, which was produced by insufficient quality and amount of data, and the statistics. Once such two slips meet at once, what will be happening? Let's suppose that a crime has occurred in some area, if AI led to a conclusion of a potential suspect, with much racial prejudice. Moreover, if the potential suspect were specified by the AI facial system, with is its low distinction image recognition, what will be happening? False choices will be systematically made again and again, based on the set biased AI crime analysis. This is a typical AI ethical problem of algorithmic bias. Most people easily think that an AI solution is worst trusting, but in reality, possibly, various fatal errors I'm talking about here are systematically produced by racial prejudice inherently kept in the actual society, and the low standard of maintaining databases; and Timnit Gebru was one of the researchers who pointed out this problem, elucidated its mechanism and

considered the possible solution for the problem in her study. And her research called Gender Shades shows a big difference between the precision of AI facial distinction of Caucasian male and that of black female.

05:21
Also, she wrote her article under the category of race and gender in Oxford Handbook of AI Ethics 2020. This problem is called algorithmic bias, but the starting point of the problem lies in the data set which includes prejudices. However, the present AI can't distinguish the data issue from the algorithm issue. And for this reason, this particular problem is not so easy to resolve. Timnit emphasizes the importance of preserving the original data which gave rise to the algorithm and keeping it under the condition that re-examinations are possible. AI shouldn't copy or reproduce unconscious prejudice, discrimination and hate which lie in the societies we live in. Also, if distorted information was sent to our society with specific interests, our understanding would be influenced by that distorted information. For an example of this, the propaganda of Nazi Germany from 1933 to 45, is the worst ruinous example in whole human history. Similarly a big data, which reflects the specific interest we're taking in AI algorithm, the society would be migrated with the abuse of AI calculation, AI calculation between quotation as its result. How then can we prevent and overcome such an algorithmic error? Today, I'd like to propose a possible scenario of solutions for this difficult problem from the Far Eastern Island country Japan. One of the key ideas I'd like to discuss can be found in Japanese Zen Buddhism, by which Steve Jobs got fascinated. Steve Jobs passed away in the year 2011 at the age of 56. So he knew neither Google get in 2012, nor the third wave AI boom after that. But if Steve were alive today, during the pandemic of COVID-19, what kind of business and what kind of AI ethics would he be working on and promoting? There is a book titled "Zen and the Art of Archery", which Steve Jobs loved to read throughout his life, written in 1936 by a German philosopher, Eugene Herrigel, born in 1884, and passed away in 1955. Originally in German, "Zen in der Kunst des Bogenschießens" this book explores the intersection between Japanese archery and the Japanese Zen practice. During the 1920s, Dr. Herrigel, who taught German idealisms philosophy in a Japanese University, namely Tohoku Imperial University, learned ancient Japanese archery or Kyūdō, not Judo, but Kyūdō, with a grandmaster Kenzô, Awa, Awa Kenzô. The word Kyūdō, which remains the way of archery from the standpoint of European rationals thought archery is originally used as a weapon which hits its target and now it has become one of the popular sports. But in Japan, still until the end of the 20th century, Zen priests practiced shooting a bow as the means of prayer. For Japanese people archery has been regarded as something quite spiritual. In this regard, let me briefly talk about Zen and Buddhism. But in order to do this, here, I'd like to introduce Dr. Everett Masaki Matsubara. He is a Zen priest and also he is working at Cornell University, Brown University, and also University of Tokyo as a visiting professor. Furthermore, he has served as a mindfulness mentor at Google. The company who also increase the G pose stress reduction system there.

09:55
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Life is suffering. We are living in tough daily life. Our life is tough, that is a Buddhist idea to think about, in other words, how it is possible to understand life is - that is the core idea in Buddhism. For example, Buddhism give us as a solution. The solution is called concept of Four Noble Truths, Four Noble Nruths. One is, obviously, there is suffering. Second, that suffering has a cause. So, there is the liberation from the suffering that you call the Nirvana. Number four, which is the last, is that there is a way to achieve the liberation, Nirvana.

11:07

Look at an image of Requiem shot by a priest. This archery place was established in the year 1977 by Reverend Cohn suhara the abbot of K show temple, located as one of the subject well the main temple called ng koozie in Kamakura. By the way in which the temple is the place where Dr. Maduro was born. The bow, Dr. Herrigel used is now stored in this archery place at ... The great master Awa taught to Dr. Herrigel in the context of Japanese archery, shooting a bow not in technique, but in spirit. In other words, the great master taught to the doctor - "don't think and don't aim the target". But why does Japanese archery teach not to think and not aim at the target?

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Why does Japanese authorities not think and not aim at a target? Do you have a better understanding of this seemingly productive statement? Let's listen to the lecture by Reverend Dr. Masaki Matsubara again.

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This is a glass of water, and then I put dirt into it. Important thing is that this glass of water represents our mind. Now I stir this glass of water now it is impossible to identify what the contents are. Again, this glass of water symbolizes our mind, the state of our mind. Exactly. We don't know. We are always wondering who we are, who we really are. So, Zen tradition tells us how it is possible to clarify what the contents are. The tradition's answer is just to place this glass of water on a flat place, and then just wait. Be patient. That is a tradition's answer to that question. After seconds, after minutes, we are getting to know what's the content, the heaviest things going down, ligther things going up, water is getting clearer. This day, there are two stages of meditation. To place this cup of water on the floor, then everything calm down. That is first stage, first phase of meditation. The second stage of meditation is the situation where we can find something new - the something new, I couldn't recognize when had picked up the dirt before putting into this cup of water. Oh, I didn't recognize a small bug here. I didn't recognize I picked up accidentally a piece of moss, or leaves. To find something new - that is a second phase of meditation. But please remember the meaning of Zen. The meaning of the Zen is simply to place this cup of water on flat place. That simple action is Zen itself.

16:13

But teachings, including Zen, teaches the importance of the way to see a thing as it is trying to avoid any prejudice. This practice is called right mindfulness, according to Zen Buddhism, when you shoot a bow the very process of thinking of the target or aiming at them is regarded as an unnecessary superfluous noise, that's to say an obstacle in attempt of the right mindfulness. What we learned from the Japanese kyudo archery is actually completely different from Western rational thinking. Why does an archer shouldn't aim at the target at glass? It's seemingly irrational idea but actually, it has also a rational background. Ancient Japanese archery is simple and light in a structure. It doesn't have its complicated structure unlike a Western rifle or crossbow get. This means that for a Japanese archer, extreme concentration and relaxation of body, mind are required. During thinking human bodies subconsciously get tense. And thus it gets too hard and having this mental and physical conditions causes fluctuations in many places of the body. When this somatic fluctuation is smaller than the error range of the shot, a bow will shoot the mark. So, an archer shouldn't feel, think or aim, but simply look at the target and let the bow go. This idea has its core essentials directly connected to the principles of AI computation, a convolutional neural network CNN or Bayesian statistics, though I won't discuss further in this talk now. Related to this idea of no thinking, an ancient Japanese archery teaches another different meaning that is an Oracle or a fortune telling. Ancient Japanese people thought that while having even no instant moment of thought an archer simply draws the bow and the arrow goes after the God's will. There is a good example that is a widely known story of folding form target. And the Tale of Heike, this story is based on a historical battle occured in the year 1185 between the Heike Samurai clan, and again the Genji Samurai clan. This one is called the naval Battle of Yashima. This story shows Nasu no Yoichi, a young archery expert of the Genji did hit the fan mark set on a boat floating in the sea. The mark was set up by the Heike clan who prepared it as a Oracle to the God of War. His shooting technique was so wonderful that even the opponent Heike samurais cheered for it. One warrior of Heike began to dance on the boat. Then the general commander of the Genji clan Minamoto no Yoshitsune ordered the Yoichi to shoot the dancing warrior. All of a sudden, the warrior was killed by Yoichi's arrow and everybody became quiet. Then the battle has begun again. For his order to shoot the dancing warrior, Yoshitsune's character was criticized even by the allies. Finally Yoshitsune was forced to commit suicide in the northern end of Japan with the age of 30. Once ordered, Japanese warrior archer hits the target without any thinking. In this context, we can argue that in the spirit of archery in a Bushido, that the way of warrior is quite similar to AI or machine learning system without any thinking and any hesitation. If we think that a living sniper shoots a living human as his or her target, we would easily see a variety of hesitation and psychological disturbance. But an AI system in fact, a military drone would shoot the target without any thinking and any hesitation lustrously. AI ethical problems related to the laser weapon is very important, so I'd like to leave it for another occasion.

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This photo shows the stage scenery of performance entitled 'Ocean' by the New York Merce Cunningham Dance Company. This is a post humorous work of the music director of the company, John Cage, a collaborator and old friend.

22:01
This photo shows the stage scenery of performance entitled 'Ocean' by the New York Merce Cunningham Dance Company. This is a post humorous work of the music director of the company, John Cage, a collaborator and old friend. You can see instrumental players, 180 musicians, sitting in a circle surrounding the dancing stage, and the audience. Assembly as an orchestra director in the first performance of this 'Ocean'. I've been working as a composer, as an orchestra opera conductor for the last 35 years.

22:38

My current laboratory started in 1989, since then, concurrent with my own music activities, my love has focused on the ethics of media and information based on both natural and social sciences and humanities.

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By the way, this lecture is for the New York University, The GovLab, and supported by Munich Technical University and the University of Tokyo. And 31 years ago, I shared a study orchestra conducting at the Leonard Bernstein in the same New York City, but all of a sudden Mr. Lenny had passed away. And finally, finally, I stayed in Tokyo, continued PhD candidate studies in physics, but music projects with a judgment of only did you judge Ligeti video and worked internationally as composer and conductor simultaneously. Then, a new wind blue in the year 1995 meeting with a French composer conductor, Pierre Boulez. I applied methods of physics to solve classical problems of music, most of which was proposed by Arnold Schoenberg, the composer, music theorist, and reformulated by Olivier Messiaen himself, Karlheinz Stockhausen, and many other musicians. Then I got PhD and appointed the professor of composition and conducting at the University of Tokyo in the year 1989.

24:13

My collaboration with the New York Merce Cunningham Dance Company, John Cage his posthumous work ... was also during that period. After that I had organized conducting method with Pierre Boulez, which I'll explain soon. I also collaborated with Bayreuth Festspielhaus, conducted music grammars of Richard Wagner just following the original stage setting and emotion captions, which have been the most important three dimensional characteristics and evaluated the space time physio community of acoustics by use of interferometric overlays analysis I prepared, on which I'll also explain soon.

24:57

During the 30s and 40s Wagner's music and Bayreuth head used for the propaganda of Nazi Germany. And then after the Second World War, the most severe ethical affliction was demanded, and they had the direct relationship with the current AI ethical problematics. My first collaboration with Munich Technical University was on the ethics of public propaganda and media mind control. My best friend Christoph Lütge has proposed important philosophical concept das notan princep, the notan principle, based on the rehab partners work the name of the sacred thought note and the music grammar, the ring of the nibble. Thus our research lab has focused on the fundamental study of music expression and their ethics in parallel with practical music making, and also with the background of physical and Information Sciences, ethical, serious, highly organized, autonomous system, self driving car, and ethics of AIs from the common deep root.

26:10

You may wonder what kind of relationship value between the artist Holocaust and AI ethics. But indeed, you'll see there are a vast amount of fundamental links between those two and more. In April 2021, an Irish digital artist is condemned severely by putting colors and even a smile on the Cambodian victims of Khmer Rouge. This is from the artists home base with contemporary AI technologies such minor changes are quite easy and there is no technical, no technical problems where the innumerable numbers of ethical problems aren't just "hey I'm faking of personal faces, voices or actions," are added against the his her or their or believed families, relatives or friends will. This is also from the artist's homepage, colors, death mask of Ludwig van Beethoven, how do you feel over this? Should we have opened his eyes with the aid of AI image processing? It's not easy to have the simple answer but it's technically very, very, very easy to manipulate documents in this manner. To any death mask and if any dead face or dead body, then how about the victims of Holocaust lost their lives in Aushwitz-Birkenau concentration camp? Once again, how do you think ethically the colors, emotions or even smiles to any historical documents or the fact, like this famous photo? Such manipulation is called deep fake and the faking on individual face, personal voice, action at any other document should be ethically throughout strictly discussed and operated but indeed, until now, we the human beings have no high roundtable of morality to do such problem. The global AI ethics consoles should play a proper role for the common benefit and future of all the human beings and our digitalized world.

28:27
Now, let's go back to focus on the viewpoint of the 20th century American artist, John Cage who was also deeply influenced by Zen Buddhism. John's idea of uncertainty means chance was little intention of particular individual. Similar to the case of Steve Jobs, John Cage was influenced by the Zen thoughts introduced by a Japanese Zen philosopher, Daisetsu Suzuki, from 1870 to 1966. In John's, well known piano piece, four minutes 33 seconds, a pianist is just sitting in front of his piano without playing the keyboards and he simply listens to the silence in the concert hall. The point of this piece is that if the pianist arbitrariness is made least, were made zero, to what would we listen? This is the simple application of Zen Buddhism's right mindfulness,. In the Eightfold ways to music listening, and the idea of four minutes, 33 seconds, means 273 seconds. And this number of 273 is based on the idea of thermodynamical absolute zero, which is minus 273 degrees Celsius. John always liked joking, and this shows an example of it.

29:54
While being called with the same word uncertainty when the Heisenberg's idea of uncertainty principle in quantum mechanics is totally different from John's uncertainty. Quantum mechanical uncertainty principle tells when measuring the velocity of a wave, the location can be specified to a certain point. And when they're specifying their location with a wave velocity of the wave can be measured clearly. As Heisenberg's inequality doesn't contain temperature T, this also explains a fluctuation of the wave function at the absolute zero degree. John's chance and Heisenberg's uncertainty is completely different still, thinking of the AI problems, I noticed that these two indeterminacies are strangely related at some zero degree point. While Cage's chance means avoidance of subjective arbitrariness, Heisenberg only tells the mathematical operation of physical phenomena. But in AI, even subjective prejudice, or errata would form a dataset and its algorithmic structure is calculated. Even subjective aggregate is regarded as physically fixed data set. Here, I'd like to put a name of researcher in the field of mathematical probability, Andrei Kolmogorov, the fundamental problems where AI always fluctuates between objective mathematical structure and subjective preference or prejudice. In the latter part of this talk, I'd like to think over the zero degree of prejudice once again. Now, I want to introduce another musician who thinks this problem is free will and uncertainty thoroughly with me, he's a French composer and conductor, Pierre Boulez. Pierre Boulez and I developed a method of conducting without figure together, which is known as spectro conducting. When we met for the first time and then 1985 as a musician, who has also studied mathematics, Boulez was hoping to remove any unnecessary

arbitrary gesture of conducting. Nine years later, after the experience of a medical autopsy and the application of digital markerless motion capture system, I noticed simple principle of corporal technique and that we synthesize so called angular dynamical methods of conducting with any ambiguous fears. Unfortunately, Pierre passed away in 2016. After his passing, I was invited to his IRCAM, French National Institute for the Research and Coordination in Acoustic Music, which Boulez has established, and I gave my lecture on the angular dynamics, which we too researched together. Thereafter, ... and I applied the machine language system to motion data Pierre Boulez had left and we extracted tangibly the rhytm and the articulation Pierre considered in his mind during the performance, posthumously. With appropriate emotion that left even after his lifetime AI can extract historical master's musical mind, and even unconsciousness during the performance. And of course, the similar analysis is applicable to living musicians, not only the conductor's but also instrumental players

33:35
AI can even extract one's unconscious habit in performing and it can object by such a unconscious habit. As physical motion data, we can apply this ability to improve performing style as well as to invent a new technique of rendering, for example, to play incredibly fast, while extremely correct, and even to prevent potential diseases. Moreover, this practice is also deeply related. The physical and the mental relaxation of the prayers were the most careful listening and practices almost the same, with the right mindfulness in Zen Buddhism, relaxation, careful listening to the words and the simple and the correct performance. During the COVID-19 pandemic, our Tokyo orchestra Mozard players, rehearsals and performances also have a maning of music Zen practice aim for the mental health and the stress reduction of the young musicians.